

CSE 4304-Data Structures Lab. Winter 23-24

Batch: CSE 22

Date: February 26, 2025

Target Group: All

Topic: Graph

Instructions:

- Regardless of how you finish the lab tasks, you must submit the solutions in Google Classroom. In case I forget to upload the tasks there, CR should contact me. The deadline will always be 11:59 PM on the day the lab took place.
- Task naming format: fullID_T01L01_2A.c/cpp
- If you find any issues in the problem description/test cases, comment in the Google Classroom.
- If you find any tricky test cases that I didn't include but that others might forget to handle, please comment! I'll be happy to add them.
- Use appropriate comments in your code. This will help you recall the solution easily in the future.
- Obtained marks will vary based on the efficiency of the solution.
- Do not use <bits/stdc++.h> library.
- Modified sections will be marked with **BLUE** color.
- You can use the STL stack unless it's specifically mentioned that you should use manual functions.

Group	Tasks
2A	1 2 3 4
1B	1 2 3 4
1A	1 2 5 6
2B	1 2 5 6
Assignments	2A/1B: 1A/2B:

Task 1: Represent a Graph using the Adjacency List and Adjacency Matrix

You are given a graph with V vertices and E edges. Your task is to represent the graph in two different ways:

- Adjacency List: This is a list of vertices where each vertex points to a list of vertices that are connected to it by an edge.
- Adjacency Matrix: This is a 2D matrix $V \times V$, where $\text{matrix}[i][j]$ is 1 if there is an edge from vertex i to vertex j and 0 otherwise.

Then, display the following information:

- The Adjacency List representation of the graph.
- The Adjacency Matrix representation of the graph.

Input Format:

- The first line contains two integers V and E , where V is the number of vertices and E is the number of edges in the graph.
- The next E lines each contain two integers u and v , indicating an edge from vertex u to vertex v .

Sample Input	Sample Output
5 6 1 2 1 3 2 4 3 4 3 5 4 5	Adjacency List: 1: 2 3 2: 1 4 3: 1 4 5 4: 2 3 5 5: 3 4 Adjacency Matrix: 0 1 1 0 0 1 0 0 1 0 1 0 0 1 1 0 1 1 0 1 0 0 1 1 0
6 3 1 2 2 3 4 5	Adjacency List: 1: 2 2: 1 3 3: 2 4: 5 5: 4 6: Adjacency Matrix: 0 1 0 0 0 0 1 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0

Task 2: Check if a Graph is k-Regular

You are given an undirected graph with a set of edges. Your task is to determine whether the graph is regular. A graph is considered regular if all vertices have exactly same number of edges.

Input:

- The first line contains two integers V and E representing the total number of vertices and edges respectively.
- The following E lines contain edge information (u,v) where each pair represents an edge between two vertices (with no loops or multiple edges).

Output:

- Print "Yes" if the graph is regular.
- Print "No" if the graph is not regular.

Sample Input	Sample Output	Explanation
4 4 1 2 2 3 3 4 4 1	Yes	Each vertex has exactly 2 edges: Vertex 1 has edges (1-2), (1-4) → degree = 2 Vertex 2 has edges (2-1), (2-3) → degree = 2 Vertex 3 has edges (3-2), (3-4) → degree = 2 Vertex 4 has edges (4-1), (4-3) → degree = 2 The graph is 2-regular.
5 6 1 2 2 3 3 4 4 5 5 1 2 4	No	Vertex 1 has edges (1-2), (1-5) → degree = 2 Vertex 2 has edges (2-1), (2-3), (2-4) → degree = 3 Vertex 3 has edges (3-2), (3-4) → degree = 2 Vertex 4 has edges (4-3), (4-5), (4-2) → degree = 3 Vertex 5 has edges (5-1), (5-4) → degree = 2 Not all vertices have the same degree, so the graph is not k-regular.
5 10 1 2 2 3 1 3 3 5 3 4 4 5 1 4 1 5 2 4 2 5	Yes	

Task 3: Topological Sort Using Queue Approach

You are given a Directed Acyclic Graph (DAG) with N nodes and M edges. Your task is to perform Topological Sort using the Queue approach.

The Queue approach is based on the Kahn's Algorithm for Topological Sorting. In this approach, we maintain a queue to store nodes with no incoming edges (i.e., their indegree is 0). We then remove these nodes from the graph and reduce the in-degree of their neighbors. This continues until there are no more nodes left in the queue.

Your task is to print the topologically sorted order of nodes.

If it is impossible to complete the topological sorting due to a cycle in the graph, return an empty array.

Input Format:

- The first line contains two integers N (the number of nodes) and M (the number of edges).
- The next M lines contain two integers a and b representing a directed edge from node a to node b.

Output Format:

- Print the topologically sorted order of nodes in a single line separated by spaces.
- If it is not possible to perform topological sorting due to a cycle, print an empty array [].

The graph is guaranteed to be valid (no multiple edges, no self-loops).

Sample Input	Sample Output	Explanation
6 7 3 4 3 5 0 1 0 2 4 5 1 3 2 3	0 1 2 3 4 5	The graph has multiple paths. Both [0, 1, 2, 3, 4, 5] and [0, 2, 1, 3, 4, 5] are valid topological sorts.
5 5 4 2 4 3 3 1 2 0 3 0	4 2 0 3 1	
4 4 0 1 0 2 1 3 2 3	0 1 2 3	Both [0, 1, 2, 3] and [0, 2, 1, 3] are valid topological sorts. The graph allows for two different valid orderings.
3 3 0 1 1 2 2 0	[]	The graph contains a cycle (0 → 1 → 2 → 0), which makes topological sorting impossible. The output is an empty array [].
6 3	0 1 2 3 4 5	The graph is disconnected, but the topological sorting

0 1 2 3 4 5		can still be done independently for each component. There can be multiple valid topological orderings like [0, 1, 2, 3, 4, 5] or [2, 3, 0, 1, 4, 5].
5 5 0 1 1 2 2 3 3 4 4 2	[]	The graph contains a cycle ($2 \rightarrow 3 \rightarrow 4 \rightarrow 2$), so topological sorting is not possible.
6 6 0 1 0 2 1 3 2 3 3 4 3 5	0 1 2 3 4 5	This is a larger graph with multiple valid topological sorts, and both orders are acceptable as valid outputs.

Task 4: Check if there exists a path between two nodes

You are given a directed graph with N nodes and M edges. The graph is represented by pairs of integers, where each pair (a, b) denotes a directed edge from node a to node b. Your task is to determine whether there exists a path from a source node src to a destination node dest.

Input:

- The first line contains two integers N and M, where N is the number of nodes, and M is the number of edges.
- The next M lines each contain two integers a and b, denoting a directed edge from node a to node b.
- The last line contains two integers src and dest, representing the source and destination nodes.

Output:

- Print "YES" if there exists a path from src to dest.
- Print "NO" if there does not exist a path from src to dest.

Sample Input	Sample Output	Explanation
4 3 1 2 2 3 3 4 1 4	Yes	There is a path from node 1 to node 4: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$. Therefore, the output is "YES".
5 8 1 2 1 3 2 4 2 5 4 4 5 2 4 3 3 5 5 3	Yes	5 2 4 3
5 8 1 2 1 3 2 4 2 5 4 4 5 2 4 3 3 5 2 1	No	
5 4 1 2 2 3 4 5 4 2 1 5	No	Although node 1 can reach 2, 3, and 4, there is no path from node 1 to node 5.
5 2 1 2	No	Disconnected Graph

3 4 1 5		
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Task 5: CG Supremacy

Aziz is working hard to improve his grades in the Data Structure course, but despite his efforts, he finds himself struggling to perform as well as his peers. While it's somewhat understandable that students with higher CG than him might make fun of him, what truly bothers him is that students with lower CGs than his are also bullying him. This seems completely unjustifiable to him.

To get to the bottom of this, **Aziz** has sought your help since he's too busy completing his pending lab tasks. Your goal is to implement **Topological Sort** algorithm to figure out who can bully him, and who he can bully in return.

Input:

- The first line contains two integers V, E and Q representing the total number of students, CG relations and Queries respectively.
- The next line will contain V strings representing student names.
- The following E lines contain two names(strings) *student1* and *student2* which represents that *student2* has higher CG than *student1*.
- The following Q lines contains a pair of student names : *student1* and *student2*

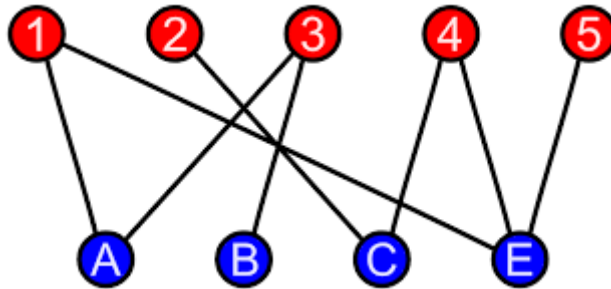
Output:

- If a valid sequence of V names exists such that each student has a higher CG than all other students before them, print the sequence (not unique). Otherwise print 'NOT POSSIBLE'.
- If such a sequence exists, print the answers to the Q queries separated by a newline where you will print **YES** if *student1* can make fun of *student2*. Otherwise print **NO**.

Sample Input	Sample Output
8 10 4 Alice Eve Frank Bob Charlie David George Hannah Alice Eve Bob David Charlie Eve David Alice David Charlie George Bob George Frank Frank David Frank Hannah Hannah Charlie George David Eve Frank Bob Charlie Alice George	George Bob Frank David Hannah Charlie Alice Eve NO YES NO YES
6 6 0 Alice Eve Frank Bob Charlie David Alice Charlie	IMPOSSIBLE

Task 6: Bipartite Graph

A bipartite graph (or bigraph) is a graph whose vertices can be divided into two disjoint and independent sets U and V , that is, every edge connects a vertex in U to one in V .



There is an **undirected** graph with n nodes, where each node is numbered between 0 and $n - 1$. The graph won't have any self or parallel edges. Given the adjacency list, determine whether the given graph is bipartite or not.

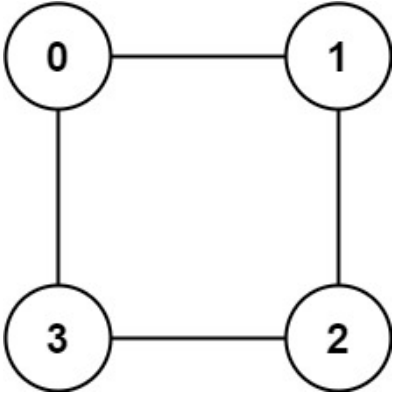
Input:

- The first line will contain a number V which denotes the number of vertices in the graph.
- The next V lines will start with a number N denoting how many vertices are connected to it followed by N vertices.

Output:

- YES if the graph is bipartite.
- NO otherwise

Sample Input	Sample Output	
4 3 1 2 3 2 0 2 3 0 1 3 2 0 2	NO	

4 2 1 3 2 0 2 2 1 3 2 0 2	YES	
4 2 1 2 2 0 2 2 0 1 1 2	NO	
9 1 1 2 0 2 2 1 3 3 2 4 6 2 3 5 2 4 8 2 3 7 2 6 8 2 5 7	YES	
6 1 1 3 0 2 5 2 1 3 2 2 4 2 3 5 2 1 4	NO	

Task 7: Budget Friendly Caraxes in Budget Friendly Westeros

In the budget-friendly version of Westeros, King (Consort) Daemon Targaryen is set on a journey riding his dragon Caraxes. However, in this version, Caraxes doesn't fly; instead, it hops across cities in Westeros. Each time Caraxes hops, it can hop over exactly k cities in one move. Given a starting location and an ending location, your task is to determine if Daemon can reach his destination and also print the path he must follow to get there, if possible.

Input:

- The first line contains two integers V , E and Q representing the total number of cities, edge information and Queries respectively.(undirected)
- The next line will contain V strings representing names of cities in Westeros.
- The following E lines contain two city names (strings) $city1$ and $city2$ which represents an edge from $city1$ to $city2$.
- The following Q lines contain src , dst and k which represents source city, destination city and number of cities Caraxes cover in each hop respectively.

Output:

- If it is possible to reach the destination city from the source, print the path.
- Otherwise print **NO**.

Sample Input	Sample Output
12 10 KingsLanding Blackwood Winterfell Dorne Dragonstone Wall Essos Lannisport Harrenhal Riverrun RedKeep StormsEnd	
KingsLanding Blackwood KingsLanding Winterfell KingsLanding Dorne Winterfell Wall Winterfell Dragonstone Dragonstone Harrenhll Dragonstone Riverrun Dorne Essos Dorne Lannisport Essos RedKeep	
KingsLanding Harrenhal 1	KingsLanding Winterfell Dragonstone Harrenhal
KingsLanding Harrenhal 3	NO
Harrenhal RedKeep 2	Harrenhal Winterfell Dorne RedKeep
RedKeep Riverrun 3	RedKeep KingsLanding Riverrun
StormsEnd Winterfell 1	NO
Essos Dragonstone 5	Essos Dragonstone